Generating Optical Illusions from Arbitrary Prompts with Diffusion Models

Xuanxin Chen, Taoyi Dai, Yinuo Li
School of Information Science and Technology, ShanghaiTech University, China
{chenxx2024, daity2024, liyn12024}@shanghaitech.edu.cn

Nisong Dong
School of Physical Science and Technology, ShanghaiTech University, China
dongns2024@shanghaitech.edu.cn

Liqin Yang
School of Biomedical Engineering, ShanghaiTech University, China
yanglq2024@shanghaitech.edu.cn

Abstract

We present a generalized framework for synthesizing multi-view optical illusions—images that reveal different semantic content under view-specific transformations such as flips, rotations, and pixel-level permutations. Extending the Visual Anagrams paradigm, our method leverages off-the-shelf text-to-image diffusion models in a zero-shot manner, allowing flexible prompt inputs and transformation types across two-view, three-view, and four-view settings. Through extensive experiments, we identify a perceptual bottleneck: illusion quality—measured by CLIP-based alignment and concealment scores—deteriorates when the number of views exceeds four. To address this, we propose a set of adaptive generation strategies tailored for the four-view regime. In particular, our Adaptive Soft Guidance method consistently improves visual alignment and semantic concealment, outperforming baseline approaches across multiple test cases. Our contributions include (1) a scalable pipeline for prompt-to-illusion synthesis, (2) empirical analysis establishing an upper bound on perceptually stable views, and (3) targeted enhancements for complex view configurations. Qualitative and quantitative results confirm the effectiveness and generality of our approach in generating coherent and visually compelling illusions under diverse transformation regimes.

1. Introduction

Visual illusions that alter interpretation under transformations—such as flipping, rotation, or spatial permutations—have captivated both scientific researchers and visual artists for over a century. From Dalí's surrealist paintings to Escher's paradoxical architectures, these illusions demonstrate how perceptual meaning can shift dramatically based on spatial context. Reproducing such effects computationally has traditionally required handcrafted designs or perceptual modeling. However, recent findings suggest that diffusion models may possess the emergent ability to mimic perceptual ambiguity, even without explicit encoding of human visual principles.

In this paper, we explore whether text-to-image diffusion models can be repurposed to generate *multi-view optical illusions* purely from textual prompts—supporting not only traditional two-view or three-view settings but also more complex four-view scenarios. Our goal is to create images that yield multiple consistent interpretations under specific transformations, without modifying the underlying model architecture. To achieve this, we build on a generalizable denoising strategy. At each diffusion timestep, the latent image is passed through a set of orthogonal geometric transformations to produce multiple view-specific versions. Each transformed view is then independently guided by its corresponding text prompt, producing distinct noise estimates. These predictions are inverse-mapped back to a shared coordinate space and combined to update the latent representation. As illustrated in Figure 1, this process enables the model to preserve spatial consistency while encoding diverse semantics across views.

Nevertheless, as the number of views increases, maintaining perceptual fidelity becomes more difficult. Our empirical analysis reveals a critical threshold at four views—beyond which illusions often exhibit structural artifacts, semantic drift, or reduced concealment. To overcome these limitations, we propose several adaptive sampling techniques tailored for the four-view case, including

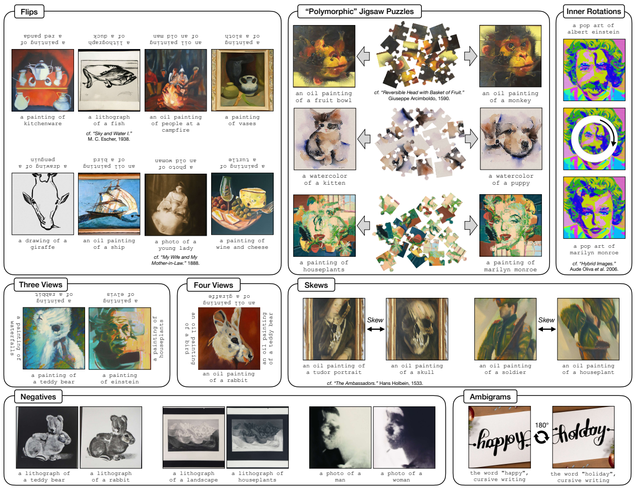


Figure 1. Our approach supports a variety of transformations, including flips, rotations, skews, color inversions, and jigsaw rearrangements.

prompt-level noise injection, CLIP-guided view filtering, and dynamically modulated classifier-free guidance.

Our contributions are summarized as follows:

- We introduce a unified diffusion-based framework for generating multi-view optical illusions from arbitrary prompts, supporting both static images and animations.
- We empirically determine that four views represent the upper bound for maintaining coherent illusions under current model constraints.
- We design a suite of adaptive enhancement techniques to improve semantic alignment and perceptual concealment in the four-view regime, including prompt perturbation, embedding fusion, and dynamic guidance scaling.
- We provide insights into compatible transformation sets that preserve both statistical assumptions and perceptual ambiguity across views.

This work provides a new perspective on leveraging pretrained diffusion models for perceptually grounded image synthesis, expanding their application to structured visual ambiguity and multi-perspective illusion generation.

Author Contributions. The individual contributions of the authors are detailed as follows:

Xuanxin Chen (35%): Led project management and task coordination. Designed and implemented algorithmic enhancements, conducted extensive experiments, and authored the *Experiments* section. Additionally, performed comprehensive manuscript revisions, refined the presentation slides, and served as the primary presenter.

Taoyi Dai (16%): Designed the presentation slides and wrote the *Motivation* section. Contributed to the overall manuscript integration and structural coherence.

Yinuo Li (16%): Composed the *Methodology* section and contributed to the refinement of algorithmic content and technical writing.

Nisong Dong (16.5%): Drafted the *Introduction* and *Background* sections, and assisted in content editing and academic language polishing.

Liqin Yang (16.5%): Wrote the *Related Work* and *Conclusion* sections, and contributed to linguistic consistency and overall clarity throughout the paper.

2. Background and Motivation

The generation of *multi-view optical illusions*—images that alter their semantic interpretation under geometric or photometric transformations such as flipping, rotation, or pixel permutation—has long intrigued both cognitive scientists and visual artists. Classic examples, such as the duck-rabbit illusion, reveal how a single visual stimulus can evoke multiple plausible interpretations, depending on context or viewpoint. These illusions exploit the brain's tendency to resolve visual ambiguity based on learned priors and are often crafted to align multiple objects in a spatially entangled yet perceptually separable manner.

Traditionally, the synthesis of such illusions has relied on handcrafted designs or perceptual modeling, limiting both their scalability and diversity. With the advent of powerful generative models—especially diffusion-based text-to-image architectures—a new opportunity emerges to automate illusion creation without explicit perceptual rules. Diffusion models, trained on large-scale image-text datasets, implicitly encode visual structures aligned with human perception, making them ideal candidates for synthesizing images that challenge or exploit perceptual biases.

However, existing attempts at illusion synthesis using diffusion models often focus on narrow transformation types (e.g., rotation) or suffer from latent-space artifacts, particularly when using architectures such as Stable Diffusion. To overcome these constraints, we explore a zero-shot framework built on off-the-shelf, pixel-space diffusion models (e.g., DeepFloyd IF), enabling coherent multi-view illusion generation through customized transformation pipelines.

Our central insight is that a single image can encode multiple semantically distinct concepts—referred to as *visual anagrams*—by fusing denoising signals from differently transformed views. This is possible due to the preservation of Gaussian noise statistics under orthogonal transformations, which maintains diffusion stability. By aligning each view with a user-defined prompt and transformation (e.g., jigsaw shuffle, patch inversion, rotation), we construct a unified generation process that flexibly supports perceptual multistability. This approach not only generalizes the classical illusion paradigm to arbitrary transformation-prompt pairs, but also opens the door to scalable and customizable generation of perceptually rich visual illusions, with applications in human vision research, adversarial design, and AI-driven creativity.

3. Methodology

Our approach builds upon the foundation laid by Geng et al. [9], who proposed a zero-shot method for generating multi-view optical illusions using diffusion models. While their work demonstrated impressive results, we identify several areas for improvement, particularly in the flexibility of transformations and the quality of generated illusions. Below, we present our unified method, which combines the strengths of their approach with novel enhancements.

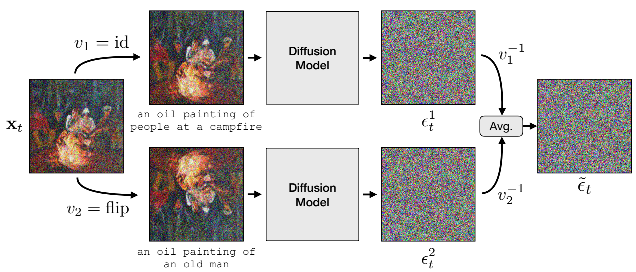


Figure 2. **Algorithm Overview.** This method works by simultaneously denoising multiple views of an image. Given a noisy image $\mathbf{x}_t$, we compute noise estimates, ϵ_t^i, conditioned on different prompts, after applying views v_i. We then apply the inverse view v_i^{-1} to align estimates, average the estimates, and perform a reverse diffusion step. The final output is an optical illusion.

3.1. Diffusion Models and Multi-View Denoising

We adopt the text-conditioned diffusion framework, where a pretrained model iteratively denoises an image $\mathbf{x}_t$ at timestep t to produce a sample $\mathbf{x}_0$ from the target distribution. The noise estimate $\epsilon_\theta(\mathbf{x}_t, y, t)$ is conditioned on a text prompt y, and the denoising process follows standard update rules such as DDPM [12] or DDIM [26]. To generate multi-view illusions, we extend this framework by simultaneously denoising multiple views of an image, each associated with a distinct prompt.

Given a set of N prompts $\{y_i\}_{i=1}^N$ and corresponding view transformations $\{v_i\}_{i=1}^N$, we compute the combined noise estimate as:

$$\hat{\epsilon}_t = \frac{1}{N} \sum_{i=1}^{N} v_i^{-1} \left(\epsilon_\theta(v_i(\mathbf{x}_t), y_i, t) \right).$$

Here, v_i applies a transformation (e.g., rotation, flip, or pixel permutation) to the noisy image $\mathbf{x}_t$, and v_i^{-1} inverts the transformation to align the noise estimates. This approach ensures that the generated image aligns with each prompt when viewed under its respective transformation.

3.2. Enhanced View Transformations

It demonstrated that orthogonal transformations (e.g., rotations, flips, and permutations) preserve the statistics of Gaussian noise, making them suitable for multi-view denoising. We expand this set of transformations by introducing the following enhancements:

Non-Orthogonal Linear Transformations. While orthogonal transformations are theoretically sound, we explore the use of non-orthogonal linear transformations (e.g., scaling or shearing) by adjusting the noise statistics during denoising. This allows for a broader range of illusions, such as images that change appearance under non-rigid deformations.

Dynamic View Scheduling. Instead of averaging noise estimates uniformly, we introduce a dynamic weighting scheme that prioritizes certain views during specific timesteps. For example, early timesteps may focus on coarse structure, while later timesteps refine details. This is formalized as:

$$\hat{\epsilon}_t = \sum_{i=1}^{N} w_i(t) \cdot v_i^{-1} \left(\epsilon_\theta(v_i(\mathbf{x}_t), y_i, t) \right),$$

where $w_i(t)$ are time-dependent weights learned or heuristically defined.

Hybrid Permutations. We combine pixel-level permutations with patch-level transformations to create more complex illusions. For instance, a jigsaw rearrangement can be applied to patches, while individual pixels within each patch undergo a separate permutation.

3.3. Noise Estimation and Statistical Consistency

To ensure the denoising process remains stable under non-orthogonal transformations, we modify the noise estimation step. For a linear transformation $\mathbf{A}$, we adjust the noise estimate to account for the change in variance:

$$\epsilon'_\theta(\mathbf{x}_t, y, t) = \mathbf{A}^{-1} \epsilon_\theta(\mathbf{A}\mathbf{x}_t, y, t) \cdot \|\mathbf{A}\|_F,$$

where $\|\mathbf{A}\|_F$ is the Frobenius norm of $\mathbf{A}$. This adjustment preserves the statistical properties of the noise, even when $\mathbf{A}$ is not orthogonal.

3.4. Integration with Latent Diffusion Models

While it highlighted artifacts in latent diffusion models due to misaligned latent orientations, we propose a hybrid approach that leverages the efficiency of latent diffusion while mitigating these issues. Specifically, we do these:

- **Align Latent Patches**. Before applying a transformation in latent space, we align the latent patches to ensure consistent orientation. This is achieved by predicting the optimal rotation for each patch during denoising.

- **Multi-Resolution Denoising**. We combine pixel-based and latent-based denoising by applying our method at multiple resolutions. Coarse structures are denoised in latent space, while fine details are refined in pixel space.

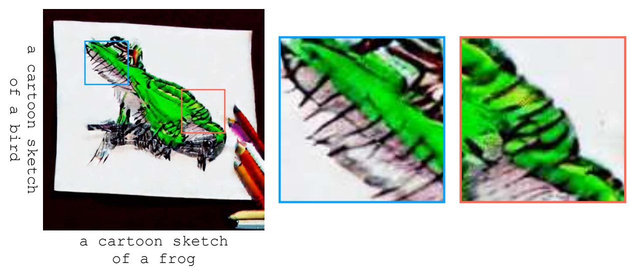


Figure 3. **Latent-Based Artifacts.** Manipulating the *location* of latent codes does not change the *orientation* of the blocks for which they encode. Therefore, when using latent diffusion models we see artifacts as shown above, in which straight lines are thatched under a rotation.

3.5. Practical Considerations

Negative Prompting. We incorporate negative prompting to discourage undesired features in specific views. For a pair of prompts (y_1, y_2), we use y_2 as a negative prompt for y_1 and vice versa, sharpening the distinction between views.

Guidance Scale Adaptation. We dynamically adjust the classifier-free guidance scale γ based on the complexity of the prompts and transformations, ensuring high-quality results across diverse illusions.

3.6. Generalized Multi-View Illusion Generation

We propose a generalized framework for generating optical illusions from arbitrary text prompts under flexible multi-view settings, building upon diffusion-based generative models. Our method utilizes the pre-trained "DeepFloyd/IF-I-L-v1.0" model from Hugging Face, integrating a T5-based text encoder to condition the generation process. To reduce memory consumption during inference, the UNet component of the pipeline is disabled, relying solely on the text encoder for semantic guidance.

A key innovation of our approach is the extension from static two-view and three-view illusions to dynamic, multi-view visual compositions. We define a rich set of view transformations—including identity, rotation (clockwise, counterclockwise, 180°), horizontal flip, negation, patch-

Figure 4. **Orthogonal Illusions.** We show that our method works, even when the view is a randomly sampled orthogonal transformation $\mathbf{A}$. While these "illusions" are incomprehensible to human perception, they serve as a confirmation for our mathematical analysis.

and pixel-level permutations, skewing, and geometric distortions—organized into pairs or sequences. These transformations form the basis for constructing complex illusionary relationships between views.

The generation process is staged in two phases: an initial sampling stage (sample_stage_1) produces a base image from the input prompt, followed by a refinement stage (sample_stage_2) that generates consistent counterparts under the specified view transformations. Throughout both stages, memory is actively managed using explicit garbage collection and GPU cache clearing.

To enhance interpretability and aesthetic appeal, we further implement various temporal interpolation functions for animating transitions between views. These animations are synthesized using the mediapy library, offering a dynamic and intuitive presentation of the underlying visual anagrams. Finally, all generated outputs and their associated metadata are systematically saved for reproducibility and downstream analysis.

4. Experiments and Improvements

4.1. Implementation

Our implementation is developed on Jupyter notebooks and is fundamentally based on the work presented in [9], which is referred to it as VA Original in the following. We have identified and rectified several issues in the original implementation, thereby enhancing its performance, particularly under conditions involving multiple views. A comprehensive comparison of our refined approach with the original version and reproduction of VA Original is also provided in https://github.com/daxuanzi515/AnyViewsIllusions-AVI.

4.1.1 Environment Setup

To meet the high RAM requirements of our project, we have chosen Google Colab and its associated computation units as our experimental environment platform. For our baseline models [2, 28], we primarily utilize the L4 schema, while our project employs the A100 schema. These schemas reflect different GPU requirements: the former necessitates

10 - 15 GB of RAM, whereas the latter requires at least 20 GB. Our improved version is based on prior work and their requirements are similar, without extra configuration.

4.1.2 Implementation Details

Adaptive Animation Module. In the original implementation, a dedicated animation module is provided to smoothly transition between generated illusion images, starting from the initial view. However, this module is limited to handling only two-view transitions. To address this limitation, we modify the original animation pipeline to support three-view sequences for reproduction purposes, and further extend it to accommodate multi-view scenarios, thereby overcoming the constraints of the original design.

Baseline Alignment. We adapt two baseline methods to ensure compatibility with our testing environment and revise their example prompts to align with our multi-view definition. Notably, IllusionDiffusion by Tancik [28] only supports the generation of two-view illusions. Although Diffusion Illusions by Burgert [2] also includes two-view generation, it incorporates additional components related to image encryption and steganography, which involve more than two prompts. Since these components are unrelated to the core task of illusion generation, including them in our evaluation would be inappropriate. Consequently, we adopt the stable two-view generation implementations for a fair comparison. Furthermore, while some baselines offer optional animation modules for visualization, others do not. To maintain fairness across all methods, we exclude these animation components from our evaluation, as they do not affect the underlying illusion generation capabilities.

4.2. Improvements

When reproducing the VA Original (Section 4.2.2), we observe that as the number of views increases, the quality of generated illusions noticeably deteriorates. In many cases, the outputs fail to reflect the intended prompts, appearing instead as incoherent color artifacts or heavily distorted images. This observation prompts us to explore the performance limitations of the original method and to identify the point at which illusion quality collapses. To this end, we propose the following research questions to guide our investigation and inform the design of our improved approach:

- **RQ1**: What is the view count threshold beyond which the VA Original method fails to generate high-quality illusions?

- **RQ2**: How can we enhance the quality of illusions near or beyond this threshold (e.g., 4-view illusions), and how effective is the proposed enhancement?

4.2.1 Metrics

We adopt CLIP-based [22] metrics to assess how well the generated illusions align with their intended prompts and simultaneously conceal alternative prompts across multiple views. Given a similarity matrix $\mathbf{S} \in \mathbb{R}^{N\times M}$ with entries:

$$S_{ij} = \phi_{\text{img}}(v_i(x))^\top \phi_{\text{text}}(p_j), \tag{1}$$

where ϕ_{img} and ϕ_{text} denote the CLIP visual and textual encoders respectively, and $v_i(x)$ represents the i-th view of the generated image x associated with prompt p_i. We define two primary evaluation metrics as follows:

Alignment Score (CLIP-A). This metric evaluates how well each view aligns with its designated prompt. We define the alignment score as the minimum value across the diagonal of $\mathbf{S}$:

$$\text{CLIP-A}_{N\times M} = \min_{i=1}^{\min(N,M)} S_{ii} \tag{2}$$

A higher value indicates better alignment between each view and its intended prompt.

Concealment Score (CLIP-C). To evaluate how well alternative prompts are concealed, we define the concealment score using the trace of row-wise and column-wise softmax of the similarity matrix:

$$\text{CLIP-C}_{N\times M} = \frac{1}{2}\left(\frac{\text{Tr}(\sigma(\mathbf{S}/\tau))}{N} + \frac{\text{Tr}(\sigma(\mathbf{S}^\top/\tau))}{M}\right) \tag{3}$$

where $\sigma(\cdot)$ is the softmax function and τ is a temperature parameter (set to $\tau = 0.01$). A higher score implies better concealment of unintended prompt content across views.

Single-View Metrics. When a single image $v(x)$ is generated from multiple prompts $\{p_1, \ldots, p_M\}$, we compute the similarity vector $\mathbf{s} \in \mathbb{R}^M$ as:

$$s_j = \phi_{\text{img}}(v(x))^\top \phi_{\text{text}}(p_j)$$

Similarly, the single-view alignment and concealment scores are defined as:

$$\text{CLIP-A}_{\text{Single}} = \min_j \mathbf{s}_j \tag{4}$$

$$\text{CLIP-C}_{\text{Single}} = \frac{1}{M}\sum_{j=1}^{M} \sigma\left(\frac{\mathbf{s}}{\tau}\right)_j \tag{5}$$

Interpretation. Both metrics are positively correlated with illusion quality. A higher **CLIP-A** score reflects stronger alignment with the intended semantics, while a higher **CLIP-C** score implies better concealment of alternative semantics. These measures are essential in assessing multi-view illusions that require both semantic specificity and ambiguity.

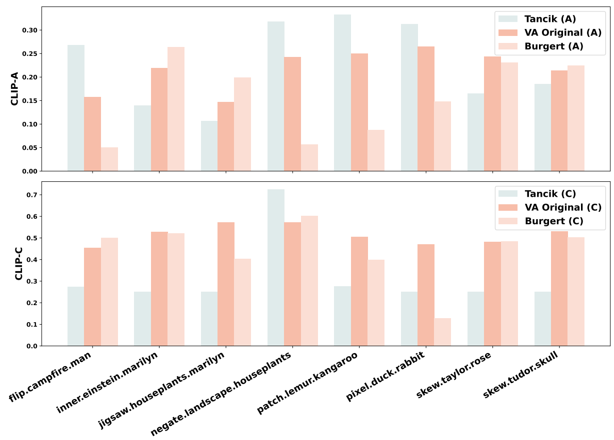


Figure 5. Comparison of CLIP-A and CLIP-C scores across eight two-views illusion examples using three baseline methods: Tancik, VA Original, and Burgert.

4.2.2 VA Original Reproduction

We begin by reproducing VA Original and two baseline methods to verify the validity of VA Original as a reliable target for further improvement. As illustrated in Figure 5, we present basic quantitative metrics to assess the illusion generation capabilities of these methods. The results indicate that Tancik achieves relatively high scores on CLIP-A in some cases but consistently underperforms on CLIP-C. Burgert, on the other hand, exhibits noticeable imbalance between CLIP-A and CLIP-C. In contrast, VA Original demonstrates stable and balanced performance across both metrics, indicating its robust and consistent ability to generate visual illusions across dual views. This validates its role as a reasonable reference point for subsequent enhancement.

Meanwhile, we also present illustrative examples in Figure 6 to showcase specific illusion images along with their descriptions. These examples demonstrate the transformation between dual-view illusions, where a single image simultaneously conveys the semantics of two distinct prompts while preserving visual completeness in both views.

4.2.3 RQ1: View Count Threshold Estimation

We observe that when the number of views increases to three or more, the quality of the generated illusions begins to degrade noticeably. To investigate this phenomenon, we conduct five examples for each view configuration to identify the critical point at which the CLIP scores begin to drop. This allows us to pinpoint the performance bottleneck of VA Original and explore potential improvement strategies. The experimental results are illustrated in Figure 7, which clearly demonstrates the inflection point at which VA Original starts to lose effectiveness. Based on the trend of CLIP scores, we identify four views as the threshold: once the number of views exceeds four, both CLIP-A and CLIP-C scores decline. Notably, for each image, CLIP-C drops more sharply, while CLIP-A remains relatively stable. This analysis reveals the scalability limitation of VA Original in multi-view settings. To address this, we propose targeted enhancement strategies in Section 4.2.4 aimed at improving its performance beyond the four-view threshold.

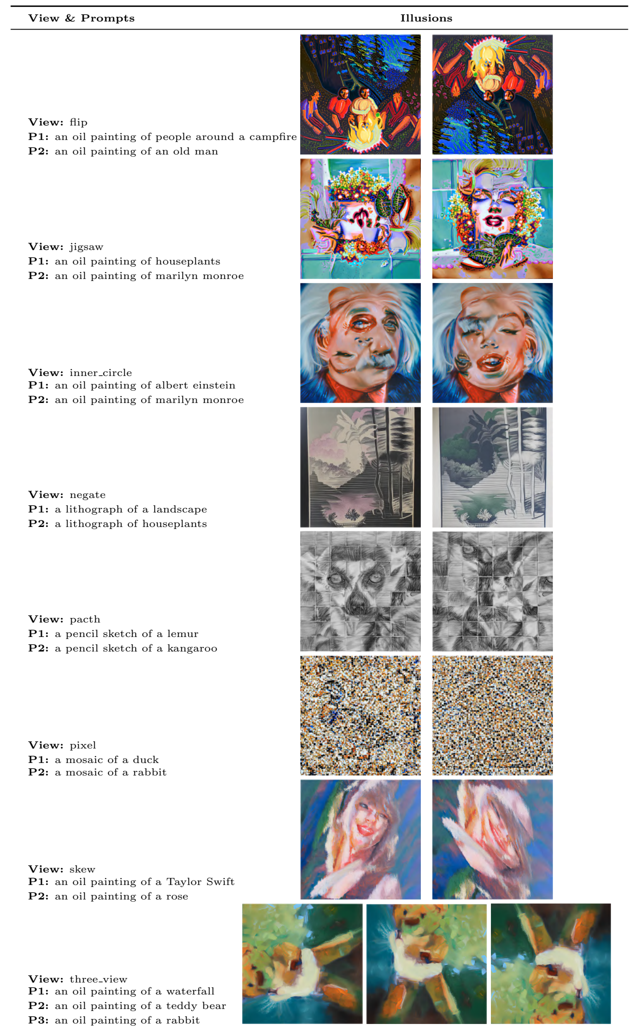
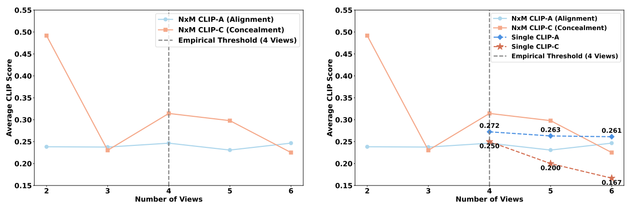


Figure 6. Illustration of visual illusions across identity and transformed views. Each example aligns a pair of prompts with a specific transformation, and the generated images show how the illusion shifts across views while maintaining conceptual alignment.

4.2.4 RQ2: Adaptive Enhancement for Threshold

According to our findings in RQ1, we observed a significant decline in illusion quality when the number of views exceeds a specific threshold. To address this limitation and enhance the robustness of multi-view generation, we propose several improvement schemas. These strategies are



(a) Threshold of 2-6 Views (b) Single CLIP of 4-6 Views

Figure 7. CLIP-A and CLIP-C score trends across different view counts. Left (a): The overall trend from two to six views shows that both $N \times M$ CLIP-A and CLIP-C scores decrease after the number of views exceeds four, indicating a threshold effect. Right (b): A focused analysis on views four to six reveals that Single CLIP-A remains relatively stable, while Single CLIP-C drops sharply—approaching zero—suggesting a significant loss in concealment as view count increases.

designed to operate under the threshold condition, where standard generation begins to degrade. Built upon the VA Original baseline, we formulate and test the following adaptive approaches:

Shared Seed Perturbation (Solution 1). This strategy introduces small-scale Gaussian noise to the prompt embeddings using a fixed random seed. It encourages controlled diversity across views while preserving global semantic consistency. By injecting prompt-level stochasticity, the method helps to mitigate mode collapse during multi-view generation.

Averaged Prompt Embedding (Solution 2). Rather than encoding each view separately, this method averages the prompt embeddings across all views and replicates the resulting representation uniformly. This enforces strong semantic alignment across views, promoting concealment at the cost of suppressing view-specific variations. It is particularly effective when the target prompts share coherent semantics.

CLIP-Guided View Filtering (Solution 3). This two-stage approach first samples preliminary images and computes their CLIP-based similarity to the text prompt. Only views with high semantic alignment are retained for final generation. Acting as a semantic quality gate, this strategy enhances robustness under ambiguous or highly variable transformations.

Adaptive Strong Guidance (Adaptive Strong). As a more assertive variant, this approach amplifies the CFG scale more aggressively according to pairwise similarity scores. While it enhances inter-view alignment, it can suppress generative diversity, leading to over-regularized results. This strategy is best suited for tasks that prioritize concealment and tight semantic consistency.

Adaptive Soft Guidance (Adaptive Soft). We propose a dynamic classifier-free guidance (CFG) modulation strategy tailored for multi-view generation, as formalized in Algorithm 1. At each denoising step t, the latent $\mathbf{x}_t$ is transformed into a set of view-specific representations $\tilde{\mathbf{X}}_t = \{v_i(\mathbf{x}_t)\}_{i=1}^N$, where v_i denotes the transformation applied to the i-th view.

The UNet predicts conditional and unconditional noise estimates, $\hat{\epsilon}^{(i)}$ and $\hat{\epsilon}_u^{(i)}$, respectively, after inverse-transforming outputs from each view back to the canonical space. We compute a view-wise similarity score α_i based on average cosine similarity with all other views, and normalize it to obtain $\tilde{\alpha}_i \in [0, 1]$.

Each view is assigned an adaptive guidance scale $s_i = \omega(1 + \lambda \cdot \tilde{\alpha}_i)$, where ω is the base CFG scale and λ controls the sensitivity to inter-view similarity. The final prediction $\hat{\epsilon}_{\text{final}}^{(i)}$ is then a weighted combination of the conditional and unconditional components. These predictions are aggregated to update the denoising process. This mechanism adaptively emphasizes semantically aligned views while preserving diversity across views, improving multi-view consistency. It proves especially effective under high view-count scenarios where fixed CFG often degrades.

Algorithm 1 Adaptive Soft Guidance

Require: $\mathbf{E}, \mathbf{E}^-, \mathcal{V}, \omega, \lambda$
Ensure: $\mathbf{x}_0$

$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$
for $t = T$ to 1 **do**
  $\tilde{\mathbf{X}}_t \leftarrow \{v_i(\mathbf{x}_t)\}_{i=1}^N$
  $\hat{\epsilon}_{\text{uncond}}, \hat{\epsilon}_{\text{text}} \leftarrow \text{UNet}(\tilde{\mathbf{X}}_t, t, [\mathbf{E}^-, \mathbf{E}])$
  $\hat{\epsilon}_u^{(i)} \leftarrow v_i^{-1}(\hat{\epsilon}_{\text{uncond}}^{(i)}), \quad \hat{\epsilon}^{(i)} \leftarrow v_i^{-1}(\hat{\epsilon}_{\text{text}}^{(i)})$
  $\alpha_i \leftarrow \frac{1}{N} \sum_{j=1}^N \cos(\hat{\epsilon}^{(i)}, \hat{\epsilon}^{(j)})$
  $\tilde{\alpha}_i \leftarrow \frac{\alpha_i - \min_j \alpha_j}{\max_j \alpha_j - \min_j \alpha_j + \epsilon}$
  $s_i \leftarrow \omega \cdot (1 + \lambda \cdot \tilde{\alpha}_i)$
  $\hat{\epsilon}_{\text{final}}^{(i)} \leftarrow \hat{\epsilon}_u^{(i)} + s_i \cdot (\hat{\epsilon}^{(i)} - \hat{\epsilon}_u^{(i)})$
  $\hat{\epsilon}_t \leftarrow \frac{1}{N} \sum_{i=1}^N \hat{\epsilon}_{\text{final}}^{(i)}$
  $\mathbf{x}_{t-1} \leftarrow \text{Scheduler.step}(\hat{\epsilon}_t, t, \mathbf{x}_t)$
end for
return $\mathbf{x}_0$

We evaluate all five proposed strategies and the VA Original baseline on nine curated four-view prompts using CLIP-A and CLIP-C metrics. As shown in Table 1, rankings based on first-place counts are used to reduce outlier effects. While Solution 3 leads CLIP-A with 5 wins, Adaptive Soft achieves the best CLIP-C performance with 4 wins—surpassing VA Original (3) and all other methods (≤ 1).

These results highlight Adaptive Soft as the most balanced strategy, offering a superior trade-off between alignment and concealment. Despite VA Original's competitiveness, Adaptive Soft consistently yields stronger concealment across diverse transformations, making it a more reliable choice for four-view illusion generation.

Test Case	A Score (Alignment)						C Score (Concealment)					
	Solution 1	Solution 2	Solution 3	VA Original	Adaptive Soft	Adaptive Strong	Solution 1	Solution 2	Solution 3	VA Original	Adaptive Soft	Adaptive Strong
rotation_objects_duck.guitar.tower.sunflower	0.2859	0.2742	0.3105	0.2721	**0.3143**	0.2228	0.2391	0.5719	0.5930	0.2355	**0.6250**	0.4698
rotation_cycle_bird.cup.cat.fox	0.2860	0.2863	**0.2697**	0.2653	0.2351	0.2149	0.2344	0.2392	0.2666	**0.6235**	0.2396	0.5329
flip_symmetry_bicycle.castle.mask.sunflower	0.2598	0.2030	**0.2720**	0.2502	0.2384	0.2228	0.2285	0.2250	0.2693	0.2257	**0.2203**	0.2368
flip_perturbation_mountain.window.balloon.house	0.2639	0.2509	0.2601	**0.3214**	0.2813	0.2477	0.2934	0.2252	0.2254	**0.6070**	0.2328	0.2545
mixed_flip_rotation_tree.train.lion.bridge	0.2268	0.2528	**0.2546**	0.2425	0.2658	0.2172	0.3986	0.2547	0.2898	**0.6222**	0.2640	0.2895
structure_destruction_fish.microwave.pool.trafficlight	0.2460	0.2986	0.2448	**0.2466**	0.2304	0.2822	0.2489	**0.3187**	0.2741	0.2699	0.2909	0.2359
style_disruptive_tornado.statue.clock.bat	0.2448	0.2183	**0.2888**	0.2118	0.2182	0.2065	0.2575	0.4602	0.4808	0.2721	**0.5639**	0.3497
low_high_freq_mix_robotdog.drone.rover.cannon	0.2719	0.2505	**0.2448**	0.2973	0.2437	0.2734	**0.2416**	0.2360	0.2775	0.2378	0.2275	0.2338
random_disruption_penguin.mirror.vacuum.speaker	0.3027	0.2322	0.2353	0.2848	**0.3377**	0.3124	0.6210	0.2289	0.2264	0.5801	**0.6250**	0.6174

Table 1. Quantitative comparison of six different methods evaluated on multi-object visual illusions under various transformation types (e.g., rotation, flip, disruption) in the View 4 setting. Higher CLIP-A indicates stronger alignment with the intended concept, while higher CLIP-C reflects better concealment of distractor interpretations. For each row (i.e., visual illusion case), the best-performing method in both CLIP-A and CLIP-C is highlighted in bold to emphasize per-example superiority.

5. Related Work

Diffusion Models. Diffusion models [5, 12, 15, 23–25, 27] have become a dominant generative framework, enabling high-fidelity synthesis across images, audio, and 3D scenes. These models gradually transform Gaussian noise into coherent data samples by learning to reverse the diffusion process. Recent advances in text-to-image generation [15, 19, 23] leverage language-conditioned denoising, allowing fine-grained semantic control. Many works have explored enhancing generation through classifier-free guidance [13], semantic guidance [17], or compositional inference [6, 16].

Computational Optical Illusions. Optical illusions have long served as a lens to probe perceptual priors in humans and machines [8, 11, 20]. Prior computational approaches include hybrid images [20], camouflage via texture retargeting [4], and generative adversarial illusions [7]. Recent works suggest that deep generative models can replicate and even extend such effects without explicit perceptual modeling [14, 18], suggesting their suitability for illusion synthesis.

Illusions with Diffusion Models. Recent studies explore using diffusion models to generate visual illusions. Tancik [29] showed that alternating noise estimates across prompts can produce perceptual shifts under rotation. Burgert *et al*. [1] applied Score Distillation Sampling (SDS) [21] to create images aligning with different prompts from different views, albeit with high computational cost. These early attempts demonstrated feasibility but lacked generality or control.

Inference with Pretrained Models. Recent work has explored modifying diffusion-based generation at inference without model fine-tuning. Prompt-to-Prompt [10] and Attend-and-Excite [3] adjust attention dynamics to enhance semantic control. The Visual Anagrams (VA Original) framework [9] fuses denoising signals from transformed views (e.g., flip, rotation) at each timestep to produce perceptually ambiguous images, but lacks support for animation and general view scalability.

We extend this line of work by introducing a generalized framework for generating optical illusions from arbitrary text prompts under flexible multi-view settings. Our system supports both static and animated outputs and provides explicit temporal control through a novel three-view animation strategy. To assess limitations in scalability, we conduct empirical studies revealing a view count threshold beyond which illusion coherence degrades. Based on this, we design five different strategies to explore which one can improve illusion qualities during threshold condition and finally provide evaluation results.

6. Conclusion

We begin by faithfully reproducing VA Original framework to validate its core mechanism in generating multi-view optical illusions. Building on this foundation, we systematically extend its capabilities in two key directions. In the Part 1 (Section 4.2.3), we explore the scalability of VA under increasing view counts and identify a perceptual threshold beyond which illusion quality deteriorates. In the Part 2 (Section 4.2.4), we investigate a series of adaptive enhancement strategies specifically tailored for the four-view configuration, which is found to be the most challenging yet viable setting.

Among the evaluated strategies, Adaptive Soft Guidance emerges as the most balanced and robust approach. It consistently achieves strong alignment with target prompts while outperforming all baselines in concealment effectiveness across diverse transformation types. This demonstrates its ability to preserve semantic fidelity without sacrificing perceptual ambiguity—a critical requirement for compelling multi-view illusions. By dynamically adjusting guidance strength based on inter-view consistency, Adaptive Soft successfully navigates the trade-off between diversity, coherence, and stealth, making it a promising solution for scaling visual illusions to more complex settings. These findings not only extend the applicability of the VA Original but also provide practical insights into designing more robust and interpretable illusion generation systems under multi-view constraints.

References

[1] Ryan Burgert, Xiang Li, Abe Leite, Kanchana Ranasinghe, and Michael Ryoo. Diffusion illusions: Hiding images in plain sight. https://ryanndagreat.github.io/Diffusion-Illusions, Mar. 2023. 8

[2] Ryan Burgert, Xiang Li, Abe Leite, Kanchana Ranasinghe, and Michael S. Ryoo. Diffusion illusions: Hiding images in plain sight, 2023. 4, 5

[3] Hila Chefer, Oran Ratzon, Amir Shalev, Lior Wolf, and Alexander Noy. Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. In *CVPR*, 2023. 8

[4] Hung-Kuo Chu, Wei-Hsin Hsu, Niloy J Mitra, Daniel Cohen-Or, Tien-Tsin Wong, and Tong-Yee Lee. Camouflage images. *ACM Trans. Graph.*, 29(4):51–1, 2010. 8

[5] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021. 8

[6] Yilun Du, Conor Durkan, Robin Strudel, Joshua B Tenenbaum, Sander Dieleman, Rob Fergus, Jascha Sohl-Dickstein, Arnaud Doucet, and Will Sussman Grathwohl. Reduce, reuse, recycle: Compositional generation with energy-based diffusion models and mcmc. In *International Conference on Machine Learning*, pages 8489–8510. PMLR, 2023. 8

[7] Gamaleldin Elsayed, Shreya Shankar, Brian Cheung, Nicolas Papernot, Alexey Kurakin, Ian Goodfellow, and Jascha Sohl-Dickstein. Adversarial examples that fool both computer vision and time-limited humans. *Advances in neural information processing systems*, 31, 2018. 8

[8] William T Freeman, Edward H Adelson, and David J Heeger. Motion without movement. *ACM Siggraph Computer Graphics*, 25(4):27–30, 1991. 8

[9] Daniel Geng, Inbum Park, and Andrew Owens. Visual anagrams: Generating multi-view optical illusions with diffusion models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3, 4, 8

[10] Amir Hertz, Ron Mokady, Tomer Tenenbaum, Kfir Aberman, Neta Perel, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control. In *ICCV*, 2023. 8

[11] Aaron Hertzmann. Visual indeterminacy in gan art. In *ACM SIGGRAPH 2020 Art Gallery*, pages 424–428. 2020. 8

[12] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *arXiv preprint arxiv:2006.11239*, 2020. 3, 8

[13] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance, 2022. 8

[14] Priyank Jaini, Kevin Clark, and Robert Geirhos. Intriguing properties of generative classifiers. *arXiv preprint arXiv:2309.16779*, 2023. 8

[15] Mikhail Konstantinov, Alex Shonenkov, Daria Bakshandaeva, and Ksenia Ivanova. If by deepfloyd lab at stabilityai, 2023. GitHub repository. 8

[16] Nan Liu, Shuang Li, Yilun Du, Antonio Torralba, and Joshua B Tenenbaum. Compositional visual generation with composable diffusion models. In *European Conference on Computer Vision*, pages 423–439. Springer, 2022. 8

[17] Xihui Liu, Dong Huk Park, Samaneh Azadi, Gong Zhang, Arman Chopikyan, Yuxiao Hu, Humphrey Shi, Anna Rohrbach, and Trevor Darrell. More control for free! image synthesis with semantic diffusion guidance. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023. 8

[18] Jerry Ngo, Swami Sankaranarayanan, and Phillip Isola. Is clip fooled by optical illusions? 2023. 8

[19] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. Glide: Towards photorealistic image generation and editing with text-guided diffusion models, 2021. 8

[20] Aude Oliva, Antonio Torralba, and Philippe G. Schyns. Hybrid images. *ACM Trans. Graph.*, 25(3):527–532, jul 2006. 8

[21] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv*, 2022. 8

[22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, pages 8748–8763. PMLR, 2021. 5

[23] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 8

[24] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S. Sara Mahdavi, Rapha Gontijo Lopes, Tim Salimans, Jonathan Ho, David J Fleet, and Mohammad Norouzi. Photorealistic text-to-image diffusion models with deep language understanding, 2022. 8

[25] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2256–2265, Lille, France, 07–09 Jul 2015. PMLR. 8

[26] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv:2010.02502*, October 2020. 3

[27] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. 8

[28] Matthew Tancik. Illusion diffusion. https://github.com/tancik/Illusion-Diffusion, 2023. Accessed: 2025-05-27. 4, 5

[29] Matthew Tancik. Illusion diffusion. https://github.com/tancik/Illusion-Diffusion, Feb. 2023. 8